

however, is approximately equal to $0.45 E_{rms}$ or $0.318 E_p$. This relationship can be seen in Fig. 1-19(c) where the average voltage is found where the shaded regions (A and B) are equal to each other. This occurs at the point equal to one-half the half-wave average voltage ($0.637 EP/2$, or $0.318 E_p$).

The half-wave rectified waveform may be unidirectional, but it is far from pure dc and thus will not be usable on many forms of dc meter movement. A measure of the departure of the waveform from pure dc (which would be represented by a horizontal line) is given by the ripple factor (1), which for half-wave sine waves is 121%.

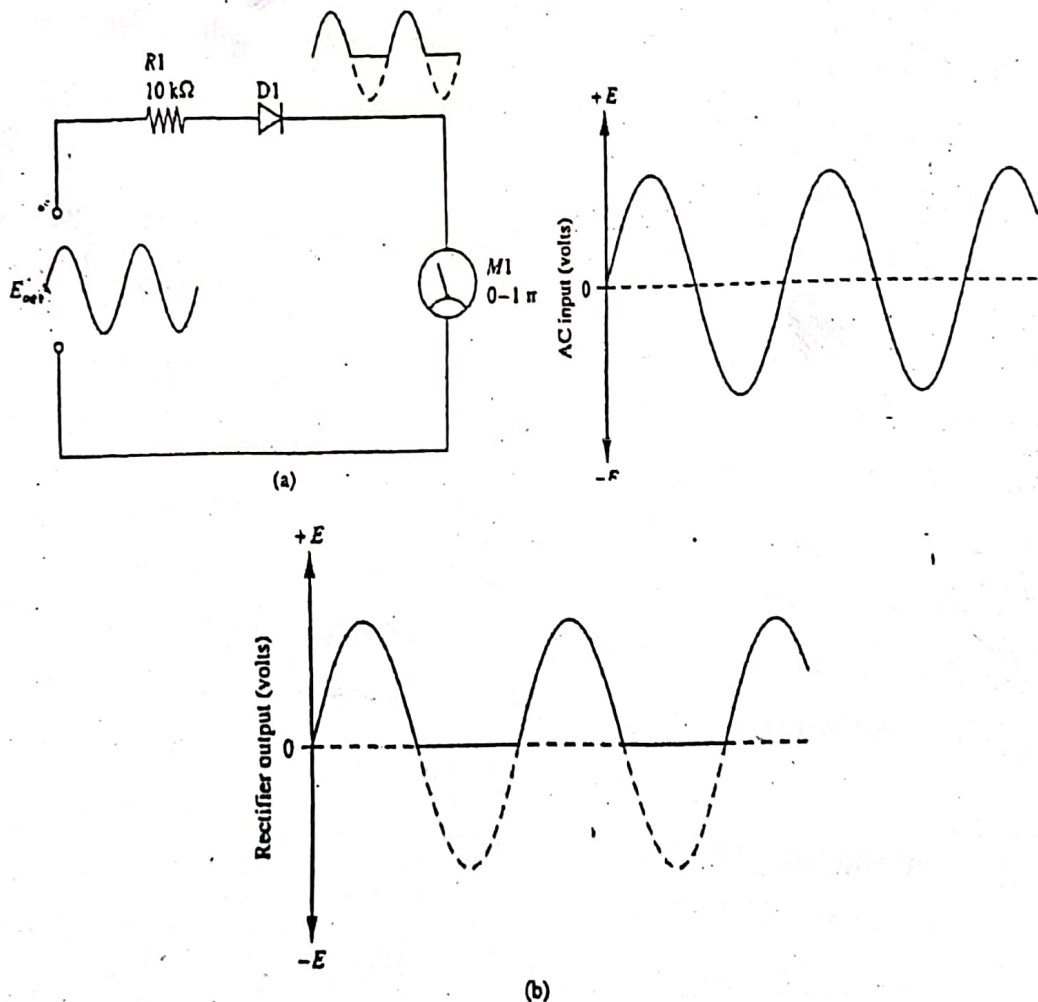


Fig. (1-19) Circuit for solid state diode, (b) applied sine wave (upper) and rectified output waveform (lower).

Pulsating dc half-wave rectified waveforms can be smoothed somewhat (i.e., made closer to pure dc) by using a ripple filter. This circuit is shown in Fig. 1-20(a). Note that the output meter $M1$ is a voltmeter shunted across a load resistor R_L and the ripple filter capacitor $C1$. The simplified explanation of how the ripple filter works is that it stores charge as the pulsating dc waveform is increasing, and then dumps the charge into the circuit as the voltage decreases. The dumped charge fills in some of the space, as shown by the shaded area in Fig. 1-20(b). As a result, the combined waveform of the pulsating dc and the charge dumped from the capacitor [the heavy line in Fig. 1-20(b)] has considerably less ripple than the original waveform, even though it is not pure dc. The value of the capacitor affects the ripple factor, with a high value producing less ripple than a lower value. Although not perfect, as a practical matter, the filtered pulsating dc waveform can be used for some ac meter applications. However, it is also true that the use of a full-wave pulsating dc waveform is even better.

The half-wave rectified waveform ignores one-half of the overall ac cycle, resulting in a value of r of 121%. *Full-wave rectification*, on the other hand, reduces ripple factor r to 48%. The lower ripple factor makes full-wave rectified pulsating dc easier to filter and easier to measure in practical circuits, especially in analog meter circuits. Figure 1-21(a) shows the ac waveform along with the full-wave rectified waveform. Note that the negative halves of the wave appear to be "flipped up" to become positive.

Although there are several forms of full-wave rectifiers ~ the one most commonly found in electronic instrumentation cases is the bridge rectifier of Fig. 1-21 (b). This circuit uses a group of four diodes arranged in a bridge circuit across the ac generator or source. Let period I be the positive half-cycle

of the input ac waveform, and period II be the negative half-cycle [see again Fig. 1-21 (a)].

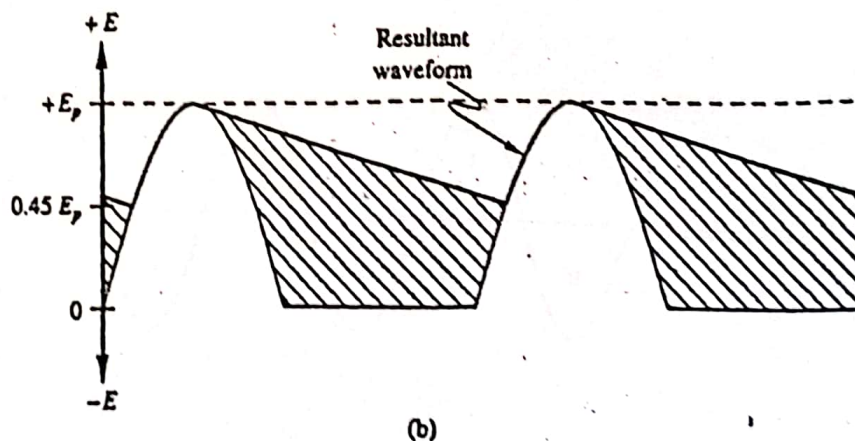
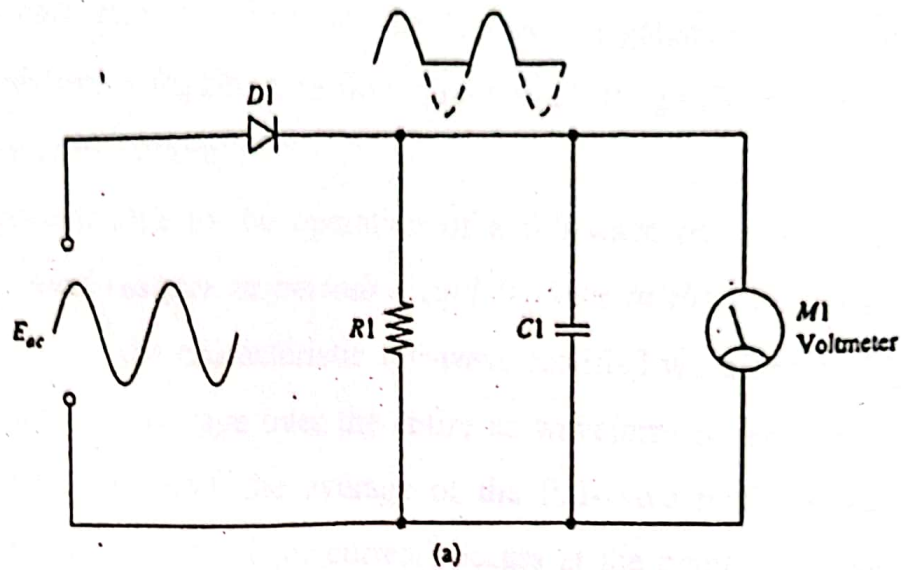


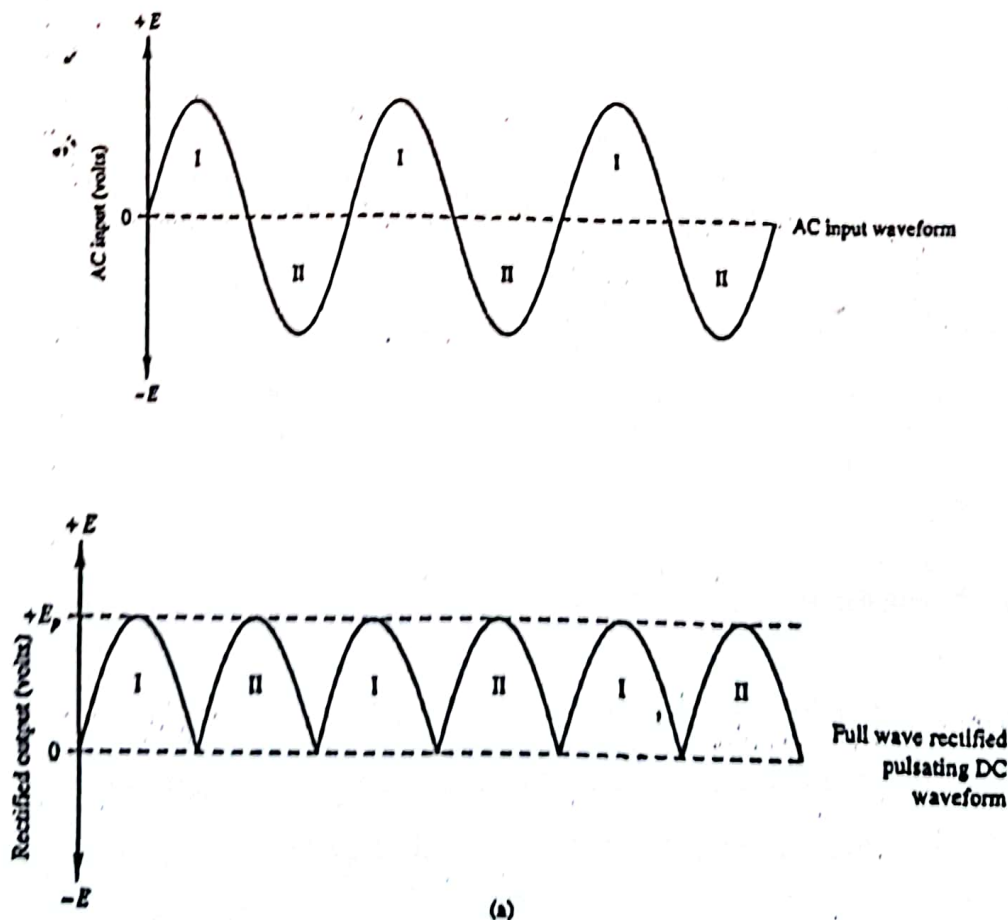
Fig. (1-20) (a) Circuit for ripple filter, (b) "brute force" ripple filter.

In phase I the polarity of the generator is such that diodes $D1$ and $D2$ are forward biased, while $D3$ and $D4$ are reverse biased. Current, flowing from negative to positive, flows out the generator, through $D2$, through load resistor R (note the flow direction!), through $D1$, and then back to the top of the generator. When the cycle reverses, making the top of the generator negative

and the bottom positive, the opposite situation occurs: diodes $D1$ and $D2$ are reverse biased while $D3$ and $D4$ are forward biased.

In this case current flows out the top of the generator, through $D3$, through load resistor R (again, note flow direction), through $D4$, and then back to the generator at its bottom.

An important clue to the operation of a full-wave rectifier is that *the current in the load resistor in periods I and II flows in the same direction*, which is what yields the characteristic full-wave rectified waveform in Fig. A :21 (a). Although the average over the entire ac waveform is zero (assuming positive/negative symmetry), the average of the full-wave rectified wave is nonzero. The average potential (or current) occurs at the point where shaded zones A and B in Fig. 1-21(c) are equal.



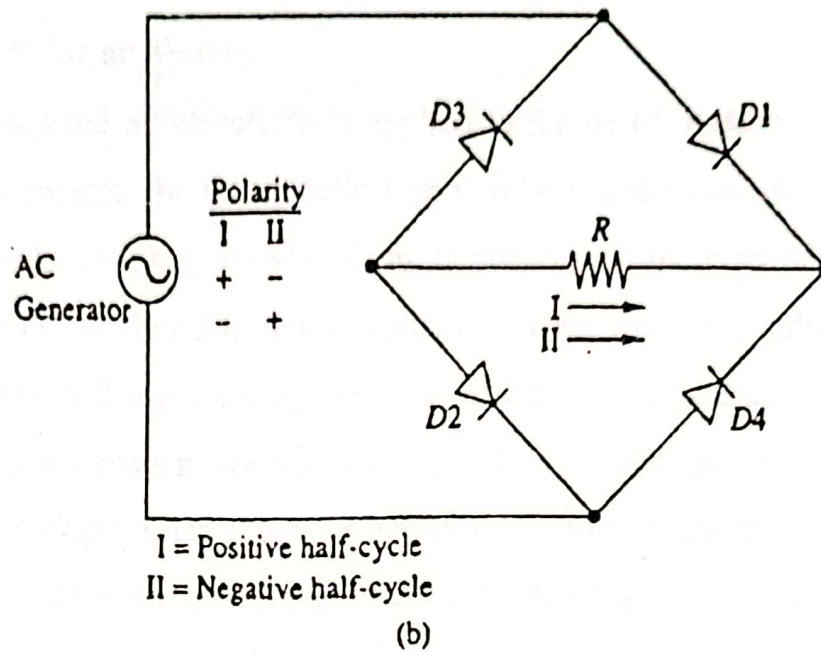
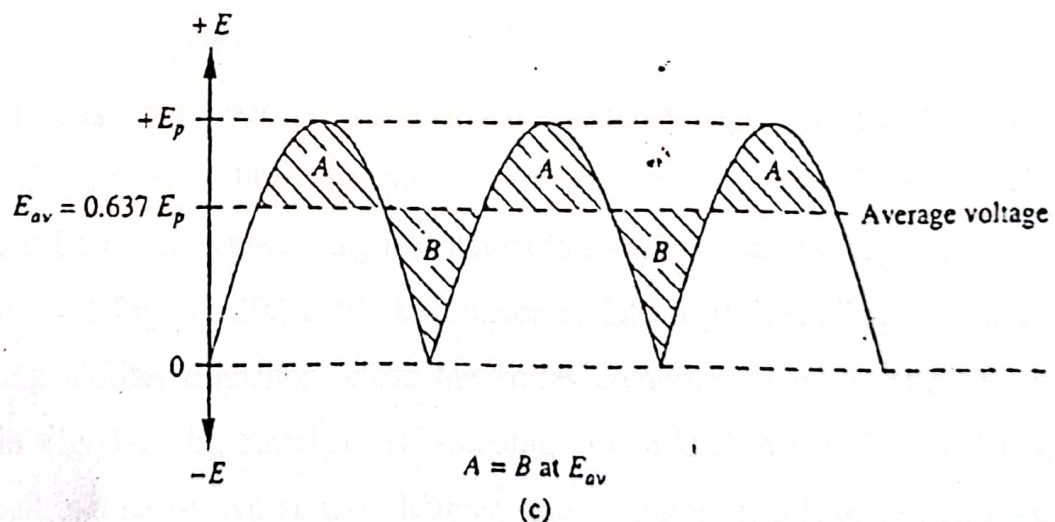


Fig. (1-21) (a) Full-wave rectification, (b) bridge rectifier circuit,

Fig. (1-21) Continued (c) Average value is the voltage at which area $A = \text{area } B$.

For a pure sine wave, this situation is such that $E_{av} = 0.637 EP = 0.9E_{rms}$. Because the half-wave symmetry of the ac waveform is not affected by full-wave rectification, the peak voltage of the full-wave rectified waveform remains at $1.411E_{rms}$.

Rectifier Circuits for ac Meters

When a rectified ac waveform is applied to the de (d' Arsonval or taut-band) meter movements, the meter deflection will be proportional to either the average or the peak values of the waveform, depending on the type of rectifier used. We will consider four different circuits: half-wave average-reading, half-wave peak-reading, full-wave average-reading, and full-wave peak-reading. The principal difference between average-reading and peak-reading instruments is the presence of a ripple filter capacitor across the output of the rectifier. The filter tends to integrate the waveform, producing a reading much closer to the peak value than to the average value.

Figure 1-22(a) shows an average-reading half-wave rectifier instrument. Although several variations on this circuit exist, this circuit is quite widely used because of its simplicity.

There is single rectifier diode in series with the signal path to the load. Resistor R_1 is used to limit the current through meter M_1 , and R_1 effectively converts it from a current-reading instrument to a voltage-reading instrument. The circuit of Fig. 1-22(a) could be converted into a peak-reading circuit by connecting a filter capacitor across the series combination of R_1 and M_1 , as shown in Fig. 1-21(b). Another peak-reading circuit is shown in Fig. 1-22(c); this circuit is a half-wave voltage doubler, and its output is a filtered pulsating dc with an average value that is close to the peak value of the unfiltered pulsating dc waveform.

Full-wave rectifier circuits are shown in Figs. 1-23(a) through 1-23(c). Two different, but related, approaches are shown in these figures: the circuit of Fig. 1-21 (a) uses a current meter indicator, while Figures 1-23(b) and 1-23(c) use a voltmeter indicator.

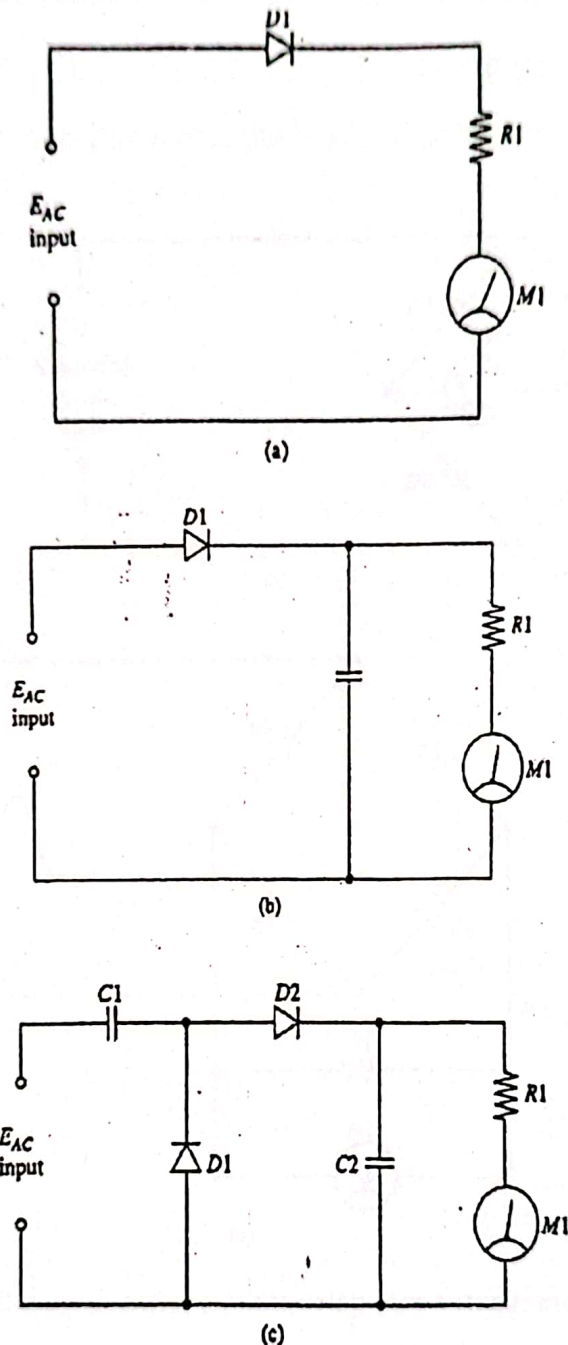


Fig. (1-22) (a) Average reading half-wave rectifier circuit; (b) peak-reading circuit with filter capacitor added; (c) half-wave voltage doubler.

Keep in mind, however, that a d' Arsonval or taut-band voltmeter is nothing but a current meter with a series resistor. The circuit in Fig. 1-23(a) is an average-reading full-wave rectified instrument because the output of the rectifier is applied directly to the meter without regard for filtering. For peak-

reading measurements, the circuit of Fig. 1-23(c) might be preferred because it has a ripple filter capacitor across the load and meter circuit.

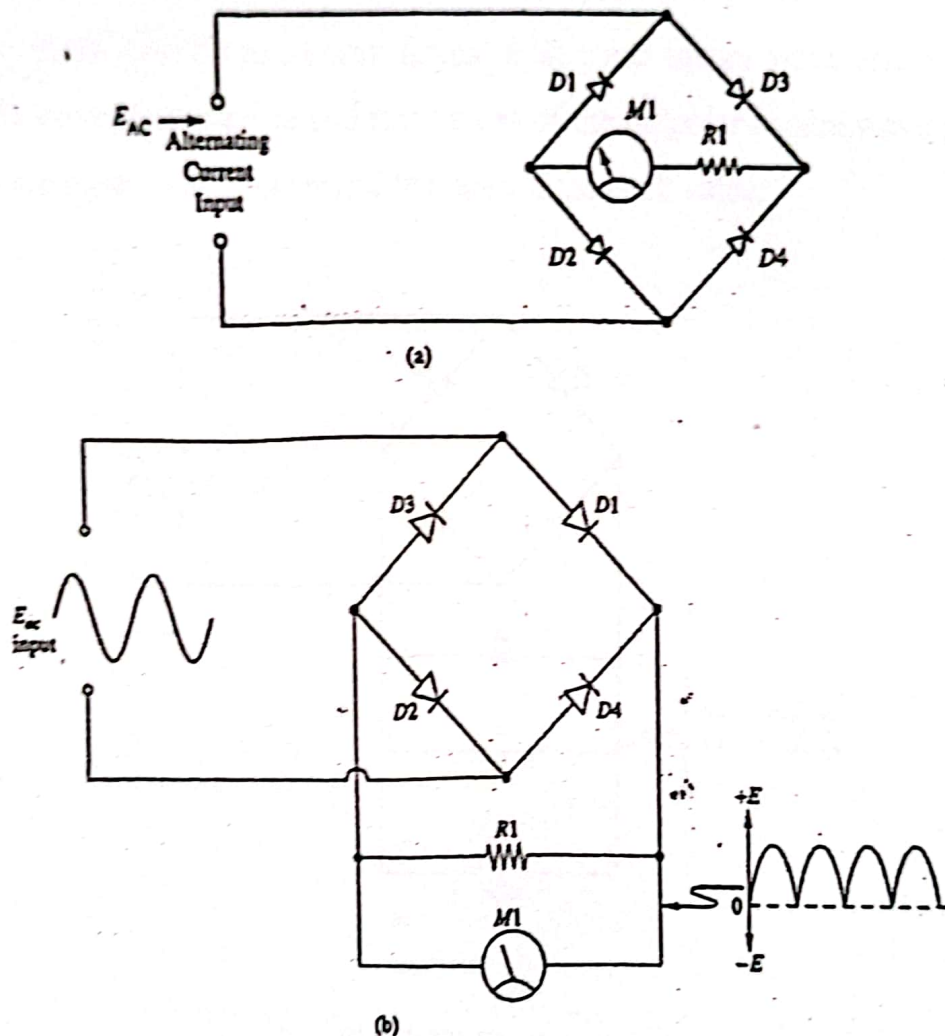


Fig. (1-23) Full-wave rectifier circuits using (a) a current meter indicator, (b) and (c) a voltmeter indicator.

Effects of Waveform on Meter Readings

Most ac meters are designed to display the rms value of the current or voltage being measured. Because the instrument measures either peak or average values, depending on the rectifier, a correction factor must be applied, and that correction factor tends to be usable only on sine waves. For an average-reading instrument, the form factor (A) is 2.22 for half-wave rectification and 1.11 for full-wave rectification. The scale will read AE_{av} . Unfortunately, A

changes with the ac waveform, so such a meter is useful only for the waveform for which it was calibrated (usually a sine wave) unless a correction factor is mentally added to or subtracted from the displayed reading. Figures 1-24(a) and 1-24(b) show two common waveforms: a unipolar square wave and a bipolar triangle wave. The average and rms values of the unipolar square wave [Fig. 1-24(a)] are equal to each other and to one-half the peak value:

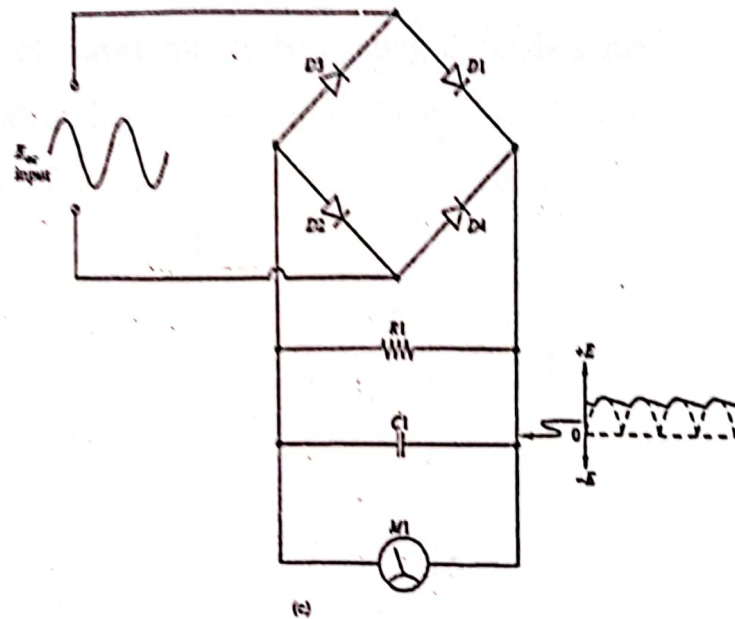


Fig. (1-23) Continued

$$E_{av} = E_{rms} = \frac{E_p}{2} \quad (1-22)$$

For the triangle waveform [Fig. 1-24(b)], the average and rms values are not equal to each other. The average value E_{av} is 0, while the rms value is

$$E_{rms} = \sqrt{\frac{E_p^2}{3}} \quad (1-23)$$

Other factors that affect the indicated value are symmetry of the input waveform across the zero baseline (j.e., is there a de offset component present?) and harmonic content of the waveform.

Meters of the sort shown in Section 3.21.2 (Figs. 1-22 and 1-23) sometimes exhibit a turnover effect; that is, they indicate a different value when the (+) and (-) leads are swapped. This problem is especially severe when the input ac waveform exhibits a lack of zero baseline symmetry

The effect of waveform on the reading depends somewhat on how the rectifier is operated. Figure 1-25 shows the E -vs.- I curve for a standard semiconductor rectifier diode. When the diode is reverse biased (-ER region), only a small leakage current ($-I_L$)

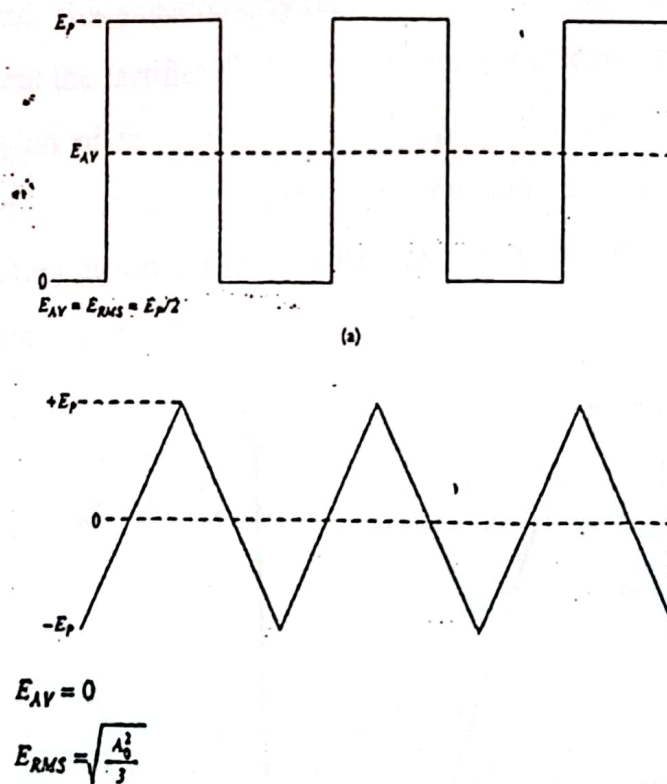


Fig. (1-24) (a) Unipolar square wave, (b) bipolar triangle wave.

Flows, which in an ideal diode would be zero. As the applied voltage goes into the forward-biased region, $+EF$, the forward current begins to rise at a nonlinear rate that is approximately equal to the square of the applied voltage. Rectifiers operated in this low voltage region are considered square law devices. Above some critical potential E_V , however, the device enters a more linear ohmic region where the forward current is directly proportional to the applied voltage. For silicon diodes, E_V is approximately 0.6 to 0.7 V.

A considerable number of waveform anomalies affect the indicated value of an ac rectifier meter. Table 1 -2 shows some of these effects. From the table it appears that, for most applications, the full-wave square law meter is the best and most consistent when nonsinusoidal ac input waveforms are being measured. This situation may require a voltage divider network on higher scales to prevent the rectifier diodes from being operated at a point outside the square law region of the device. It also suggests that peak-reading instruments be avoided unless the peak value is somehow important (transmitter amplitude modulation measurements and RF power measurements in the single-sideband mode are examples).

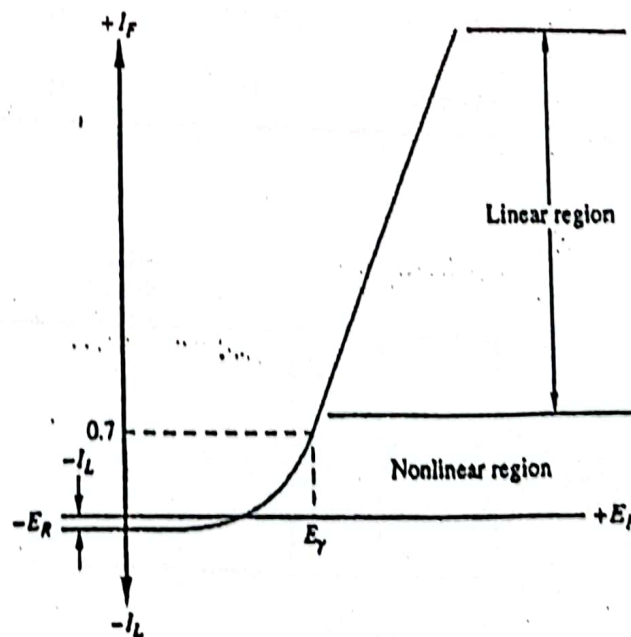


Fig. (1-25) E -vs.- I curve for standard semiconductor rectifier diode.

True rms-Reading Rectifier Instruments

A meter that reads the true rms value of an ac waveform (instead of merely adjusting the scale according to the form factor A to indicate E_{rms}) will take into account most of the anomalies discussed above. Rectifier instruments are inherently either average-reading or peak-reading devices, depending on the rectifier types. However, the use of a circuit called an rms-to-dc converter (Fig. 1-26) will produce a dc output voltage that is equivalent to the rms value of the input waveform. These circuits were once quite expensive and complex, but in modern integrated circuit form they have become quite simple from the external perspective.

Table (1-2) Effect of waveform anomalies on ac meter readings.

Effect	Full-Wave Square Law	Half-Wave Square Law	Linear	Peak Reading
Turnover effect	No	Yes	No	Yes
Phase of harmonics affect reading?	No	Yes	Yes	Yes
Effect of 2nd harmonic (%)	11	-6 to +27	0 to 10	-25 to +50
Effect of 3rd harmonic (%)	11	12.5	-10 to +16	-8 to +50

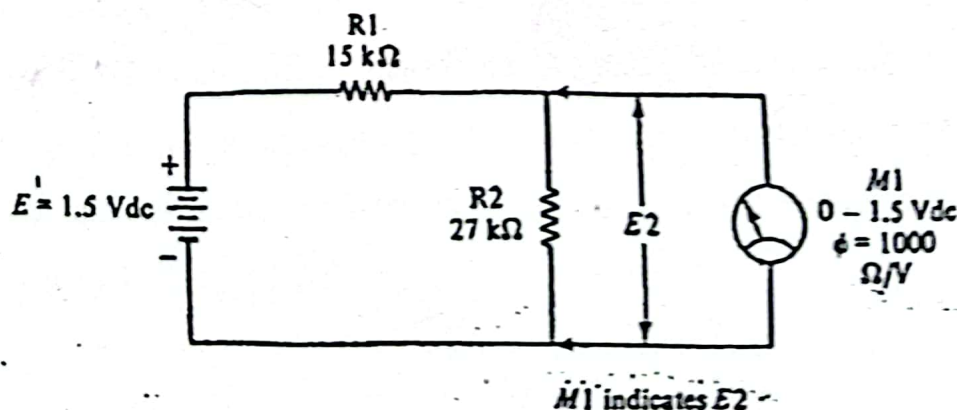


Fig. (1-26) rms-to-dc converter

Other AC-Reading Instruments

Alternating current values can also be measured on the oscilloscope, on the audio voltmeter, and on electronic multimeters.

We will not discuss it here except to state that it allows the user not only to determine the peak and peak-to-peak values of the applied waveform but also to evaluate the *wave shape* and the *period* of the waveform. Many oscilloscopes today are equipped with built-in digital voltmeters, so they can provide numerical readouts of the peak, peak-to-peak, rms, and average voltage values, as well as period and frequency information. Some models have separate displays for the numerical data, while others print it right on the CRT with the wave shape.

The audio voltmeter is a rectifier or IC rms module instrument that is preceded by a wideband amplifier. This amplifier greatly increases the sensitivity of the meter, allowing signal levels as low as -100 dBm to be measured. (Note: 0 dBm is defined as 1 mW of 1000-Hz signal dissipated in a 600- Ω resistive load.)

Electronic multimeters are instruments that contain dc voltage-, direct current-, ac voltage-, and resistance-measuring capability in a single case, with a common display or meter movement. These instruments today are almost entirely digital in operation. Earlier forms of electronic multimeters include devices that are essentially volt-ohm-millimeters (VOMs) with a field effect transistor (FET) amplifier in the front end and a balanced differential amplifier to drive the meter movement. A switchable frequency-compensated, input voltage divider network was used to reduce higher-level input voltages or currents to the native base range of the instrument.

Summary :

1. The two basic dc meter movement types are the d' Arsonval and the taut-band. Both are examples of the permanent magnet moving coil (PMMC) galvanometer.
2. The range of a current meter can be extended by using a shunt resistor.
3. A dc current meter can be used to measure voltage by connecting a multiplier resistor in series with the meter.
4. An ohmmeter can be made from a dc current meter if a precision-regulated voltage source is provided.
5. A multimeter, called a volt-ohm-milliammeter or VOM, is an instrument that has switch-selectable voltage, current, and resistance ranges.
6. There are four basic types of ac meter movement: thermocouple, hot-wire, electrodynamicometer, and iron-vane.
7. Direct current. (dc) meter movements can be used to measure ac values if a rectifier is used in the circuit.
8. Different indications are possible from different types of ac instruments used to measure the same potential difference, because some instruments read some, value, some read average value, and one reads the average of the current squared.
9. Calibration of many ac meters assumes a sinusoidal or near-sinusoidal ac waveform. A correction factor is often used when other waveforms are measured.

RECAPITULATION

Now go back and try to answer the questions at the beginning of the chapter. When you are finished, answer the questions and work the problems below. Place a mark beside each problem or question that you cannot answer, and then go back to the text and reread the appropriate section.

QUESTIONS :

1. An ammeter is connected in----- with the load.
2. A voltmeter is connected in with the load.
3. Name two types of PMMC galvanometer meter movements: -----and-----
4. The same basic electrical principles govern the behavior of the PMMC meters and
5. The range of an ammeter, milliammeter, or microammeter can be increased if a ----- resistor is used in with the meter movement.
6. A voltmeter can be made from a de current meter movement if a _____ resistor is connected in with the meter movement.
7. Draw the circuit diagram for a simple de ohmmeter.
8. A current meter should have an internal resistance that is _____ compared with external circuit resistances . .
9. A voltmeter should have an internal resistance that is _____ compared with external circuit resistances.
10. An ohmmeter may be used in circuits with the power turned on. True or false?
11. A thermocouple ac ammeter reads the value of the ac waveform.
12. Thermocouple ammeters are used at frequencies up to MHz.
13. A hot-wire ammeter reads the value of the ac waveform.
14. A is an ac ammeter constructed from a movable coil in the magnetic fields of two stationary coils. All three coils are connected in _____ with each other.
15. The ammeter in Question 14 is calibrated in rms values but is actually reading - - ---- values.
16. A is used to extend the current range of the electro dynamometer type of ac current meter.
17. Iron-vane meters read the value of the ac waveform.
18. What happens to the iron-vane meter when de is measured? Can you think of a method for overcoming this problem?
19. A rectifier meter reads the value of the alternating current or the ac voltage waveform.
20. A diode is often used as the instrumentation rectifier in audio frequency ac meters.
21. An ac meter reads different values depending on which way the meter probes are connected. This is called - -----
22. A rectifier-type ac voltmeter is operated such that the voltage applied to the rectifier diode is less than V_r This is an example of a _____ device.

23. Which type of ac meter rectifier is immune to both turnover effect and the effects of the phase of harmonics on the reading?

PROBLEMS

1. You have a 0-to-50- μ A dc current meter that has an internal resistance of 4500 Ω . What value of shunt resistor is needed to make the meter read 0 to 500 μ A full scale?
2. A dc current meter is in series with a 12- Ω load that is connected across a 9-V dc battery. How much current flows in the meter if it is assumed that both meter and battery are ideal?
3. A battery has an open-circuit (i.e., no-load) potential of 12 V dc and an internal resistance of 22 Ω . A 0-to-1-mA dc meter with an internal resistance is accidentally connected across this battery. How much current flows in the meter movement?
4. Find the value of a shunt resistor required to make a 0-to-100- μ A dc current meter read 0 to 1 μ A if the meter's internal resistance is 1100 Ω . Use (a) Ohm's law and (b) current divider methods, and compare the results.
5. Solve Problem 4 for a 0-to-20- μ A meter movement.
6. What value of multiplier resistor will make a 0-to-100 μ A meter with an internal resistance of 1300 Ω read 0 to 100 V?
7. Calculate the multiplier resistance needed to make the meter in Problem 6 read 0 to 2 V. Show your work.
8. What is the sensitivity of a voltmeter constructed from a 0-to-30- μ A dc meter movement?
9. Calculate the impedance of a dc voltmeter on the 0-to-50-V scale if the sensitivity is 20,000 Ω /V.